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Classification of Crop Plants based on Growing Season, Temperature Requirement and Photosynthetic Behavior

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Abstract

Crops can be grouped based on growing season, growing degree day (GDD), photosynthetic behaviors. Based on season, there are main four groups of crop such as autumn, winter (also known as rabi/cold season crops), summer (also known as warm season/zaid crops) and rainy (also known as kharif/ monsoon crops). Autumn season crops are grown during autumn period from September to October. They are planted in September and harvest period may last up to November-December. Winter crops are usually planted at the end of October to December and harvest period extend to February-March but most of their life cycle occurs in the winter season such as wheat, gram, sunflower, etc. Spring season crops are planted in February-March and harvest period extend to May. Summer season crops are those crops planted in March-May and harvest period may extend to June-July such as groundnut, watermelon, pumpkin, gourds. Rainy season crops are planted at the end of June to July and harvest period extend to September such as cotton, rice, jowar, bajara, etc. Off-season crops are those crops which can grow in cross season ie either season with some modification in cultivation practices and management. Day neutral crops are those crops which are insensitive to photosynthesis. These crops can be grown in all round the year. Cole crops is a general term used to describe several vegetables in the cruciferae family, including broccoli, brussels sprouts, cabbage, cauliflower, collards, kale and kohlrabi. All cole crops are the species of Brassica oleracea. They are cool-season vegetables that prefer 15° to 21°C temperatures for optimal growth and can withstand light frosts without injury. On contrary to temperature and light specific requirement of the crops, some cultivars of many crop species are developed by breeding tools as day neutral plants such as early rice (Chaite Dhan).

Keywords: Growing degree day, cole crops, CAM plants, off-season crops, day neutral plants

Introduction

Crops need optimum levels of light, temperature, moisture and other environmental conditions for optimum growth and development. When agriculture plants are classified, environmental requirements (growing season) of such crop is the main basis of classification. However, crops can be grouped based on growing season, growing degree day (GDD), photosynthetic behavior such as C3, C4 or CAM plant. Temperature is important for seed germination and plant growth. Different crops require different temperatures for germination. According to the temperature requirement of the crops, they are grouped into winter crops, spring crops, summer crops, rainy season crops, autumn crops, day neutral crops, off-season crops, etc. These classification are summarized below and number of existing species in some of these groups are given in **Figure** 1.

- **Growing season base**: Autumn crops, Winter crops, Summer crops, Spring crops, Rainy crops, Day neutral crops
- **Frost Tolerant base**: Hardy, Half hardy, Tender, Very tender
- **Photosynthetic behavior**: C3, C4, CAM

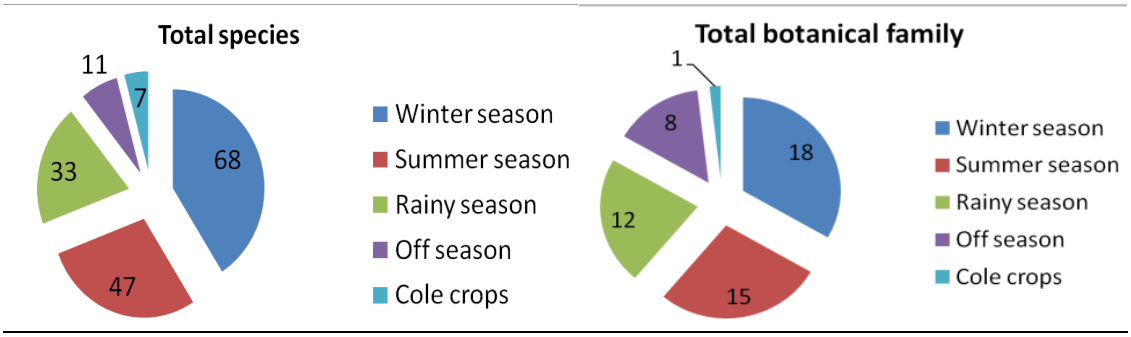


Figure 1. Number of species belonging to the growing seasons.

Autumn Crops

Autumn season crops are grown during autumn period from September and October. They are planted in September and harvested period may last up to December. Autumn season is between rainy and winter seasons during which temperatures gradually decrease. It is often called fall in the United States because leaves fall from the trees at that time. Autumn is usually defined in the Northern Hemisphere as the period between the autumnal equinox (day and night equal in length), September 22 or 23, and the winter solstice (year's shortest day), December 21 or 22. The autumn temperature transition between summer heat and winter cold occurs only in middle and high latitudes; in equatorial regions, temperatures generally vary little during the year. In Nepal autumn is shortest season.

Winter Crops

Winter crops are usually planted in October-November and harvest period extend to April but most of their life cycle occurs in the winter season. Examples of these crops are wheat, barley, lentil, mustard, most of the cole crops, faba bean, clover, fenugreek, checkpea, lupin, flax, rye grass, grasspea, etc. These crops are originated in temperate climate and have their favorable growth period during the cool parts of the year. They are mostly frost tolerant.

Spring Crops

Spring season crops are planted in February-March and harvest period extend to April-May.

Summer Crops

Summer or warm season crops are usually planted in the beginning of summer season and most of their life cycle occurs in the summer. They are often harvested in June/July. Examples of these crops are maize (corn), grain sorghum, sweet sorghum, cotton, peanut, sesame, sunflower, soybean, etc. Summer crops primarily are from subtropical and tropical regions and require warm weather for seed germination and plant growth. These crops grow better during warm temperatures. Average daily temperatures ranges from 21°C to 32°C. They are frost susceptible. They are injured or killed by freezing temperatures. Summer crops are planted in March through the middle of May. The soil temperature at the depth where the seeds are planted should be at least 10°C.

Summer Vegetables and Climate Requirements

Bean, cucumber, brinjal, okra pumpkin, pepper, squash, tomato need a minimum 16°C and optimum range of 24°C to 35°C. Many warm-season vegetables can be grown out of their season if they are protected from temperatures below 10°C. Vegetables can be protected from low temperatures with cloches, row covers, cold frames, hot bed, and greenhouses. Tender plants started under protection should not be set out before the date of the first frost. Plants started in a cold frame or under cover should be hardened gradually by exposure to the outdoor air before transplanting. With regard to success stories of crop diversification, many farmers have successfully adopted cultivation of

different off-season vegetables like cabbage, peas, cucumber, tomato, etc using modern technologies.

Warm-season crops should be planted so that they mature when the weather is warm, when the soil and air temperatures are above 10°C. They grow best when the temperature is 24°C or warmer. Warm-season vegetables can be grown out of their season if they are protected from temperatures below 10°C. Warm-season crops are often categorized as tender and very-tender.

Rainy Season Crops

Rainy season is between summer and Autumn during which most of the rainfall occur in Nepal. Rainy season crops are planted in July-August and harvest period extend to September.

Off-season Crops

Off-season crops are those crops which can grow in cross season ie either season with some modification in cultivation practices and management. For examples, some vegetables such as tomato, cauliflower, cucumber, onion can be grown and produced out of normal season. This can be accomplished by using the different agro-climatic conditions, adjusting the planting time, selecting and improving the varieties or by creating the controlled environment by making the plastic tunnels, polythene houses, permanent glasshouses, etc. Off-season vegetable is being recognized as a major agriculture commodity that holds the comparative advantage for export to India (HVAP 2011). Between July and September, the agro-climatic conditions in the Tarai of Nepal and India become unsuitable for vegetable cultivation. Due to monsoon water logging and flooding occurs in Tarai region and therefore can not cultivate vegetables during this rainy season. This potentiality can be exploited and a new export market opened up for Nepalese farmers.

Day Neutral Crops

Day neutral crops are those crops which are insensitive to photosynthesis. These crops can be grown in all round the year. Due to advance breeding technologies, nowadays, there are many crop varieties available for year round production. For example, maize can be grown in winter, summer and rainy season. Similarly some rice cultivars can be grown in winter such as Hiunde rice, and some are early season rice (Chaite Dhan).

Cole Crops

Another group of crops which are members of the cruciferae family and grown in winter are called cole crops. Cole crops is a general term used to describe several vegetables in the cruciferae family, including broccoli, brussels sprout, cabbage, cauliflower, collards, kale and kohlrabi. All cole crops are the species of *Brassica oleracea*. They are cool-season vegetables that prefer 15°C to 21°C temperatures for optimal growth and can withstand light frosts without injury. Cole crops originated from the word caulis (Latin) meaning stem or stalk of a plant. It is assumed that the cabbage and kales originated in Western Europe and cauliflower and broccoli came from the Mediterranean region. Cabbages and kales were the first to be domesticated. Prior to cultivation and use as food, they were used mainly for medicinal purposes. The cole crops are biennials but are generally grown as annuals. The brassica family is well adapted to cool season production. These plants are quite cold resistant. Young hardened cabbage plants can withstand -10°C for a short time. The growth rate of cabbage stops at 0°C and is fastest at 15°C to 20°C. Above 25°C growth stop. The minimum temperature for seed germination is 5°C with an optimum germination temperature of 27°C, an optimum range of 7°C to 27°C and a maximum germination temperature of 37°C. Cauliflower and broccoli will not stand temperatures as high or low as cabbage. High temperatures delay maturity and increase vegetative growth (number of leaves) and cool temperatures hasten maturity and may induce bolting'. Young hardened plants can withstand -5 to -10°C. The cole crops are extremely

important to vegetable production in the Nepal. They are important fresh and processing crops which are increasing in consumption. Followings are the cole crop species.

- Cabbage - *Brassica olerace var Capitata*
- Cauliflower - *Brassica oleracea var Botrytis*
- Broccoli - *Brassica oleracea var Italica*
- Brussels Sprouts - *Brassica oleracea var Gemmifera*
- Kale - *Brassica oleracea var Acephala*
- Kohlrabi - *Brassica oleracea var Gongylodes*
- Collard -*Brassica oleracea var Acephala group*
- Chinese cabbage - *Brassica chinensis L.*



Figure 2. Kale (left) and cauliflower (right).

Table 1. Crops of different growing seasons in Nepal

SN	Summer Crops	Rainy Crops	Winter Crops	Off season Crops	Cole Crops
1	Maize	Rice	Wheat	Maize	Cauliflower
2	Cowpea	Maize	Sugarcane	Cauliflower	Broccoli
3	Mungbean	Sorghum	Barley	Cabbage	Chinese cabbage
4		Bajra	Chickpea	Cucumber	Cabbage
5		Pigeon Pea	Lentil	Egg plant	Brussels sprouts
6		Finger millet	Peas	Broad leaf mustard	Kale
7		Proso millet	Rapeseed	Tomato	Collard
8		Baryard millet	Mustard	Carrot	Kohlrabi
9		Foxtail millet	Linseed	Radish	
10		Blackgram	Safflower	Soybean	
11		Mugbean	Sunflower	Summer squash	
12		Soybean	Oats	Swiss Chard	
13		Grounut	Sugar beet		
14		Sesame	Potato		
15		Cotton	Tobacco		
16		Jute	Spinach		
17		Dhaicha	Broad leaf mustard		

Table 2. List of winter crops

SN	English Name	Botanical Name	Nepali Name	Off season Habit	Family
1	Wheat, Bread wheat	Triticum aestivum L.	Gaun	No	Gramineae
2	Barley	Hordeum vulgare L.	Jau	No	Gramineae
3	Two rowed barley	Hordeum distichon L.	Dui line jau	No	Gramineae
4	Six rowed barley	Hordeum hexastichon (L.) Asch.	Chha line jau	No	Gramineae

SN	English Name	Botanical Name	Nepali Name	Off season Habit	Family
5	Naked barley	Hordeum vulgare L.var. nudum Hook. f.	Uwa	No	Gramineae
6	Tartary buckwheat	Fagopyrum tataricum (L.) Gaertn.	Tite Phaper	No	Polygonaceae
7	Common buckwheat	Fagopyrum esculetum Moench	Mithe Phapar	No	Polygonaceae
8	Wild buckwheat	Fagopyrum cymosum (Trevir.) Meisn	Jangali Phapar	No	Polygonaceae
9	Wild buckwheat	Fagopyrum megacarpum H.Hara	Jangali Phapar	No	Polygonaceae
10	Triticale	Triticale hexaploide Lart	Tritikeli	No	Gramineae
11	Fodder oats	Avena sativa L.	Jai	No	Gramineae
12	Sugarcane	Saccharum officinarum L.	Ukhu	No	Gramineae
13	Sugarcane	Saccharum complex	Ukhu	No	Gramineae
14	Sugarcane	Saccharum officinarum L.	Ukhu	No	Gramineae
15	Sugarcane	S. barberi Jeswiet	Ukhu	No	Gramineae
16	Sugarcane	S. sinense Roxb.	Ukhu	No	Gramineae
17	Sorghum	Sorghum biclor Moench	Junelo	No	Gramineae
18	Potato	Solanum tuberosum L.	Aalu	No	Solanaceae
19	Sugar beet, Beet root	Beta vulgaris L.	Chukondar	No	Chenopodiaceae
20	Lentil	Lens culinaris Medikus ssp. culinaris	Musuro	No	Leguminoceae
21	Chickpea	Cicer arietinum L.	Chana	No	Leguminoceae
22	Cluster bean	Cyamopsis tetragonalobaTaub.	Jhuppe simi	No	Leguminoceae
23	Faba (Broad) bean	Vicia faba L. var. eu faba major	Bakkulla	No	Leguminoceae
24	Field pea	Pisum sativum L. var. arvense L.	Sano kerau	No	Leguminoceae
25	Garden pea	Pisum sativum L.	Kerau	No	Leguminoceae
26	Garden vetch	Vicia sativa L.	Kutuli kosa	No	Leguminoceae
27	Grasspea	Lathyrus sativus L.	Keshari	No	Leguminoceae
28	Globe artichoke	Cynara cardunculus L. sub sp. scolymus (L.), syn. Cynara scolymus L.)		No	Asteraceae
29	Jerusalem artichoke	Helianthus tuberosus L	Gane suryamukhi	No	Asteraceae
30	Broadleaf Mustard	Brassica juncea (L.) Czern & Coss. var Rayo	Rayo	Yes	Cruciferae
31	Mustard, Rapeseed	Brassica campestris (L.) var. toria Dutch	Tori	No	Cruciferae
32	Sarson	Brassica campestris L. var. sarson Prain	Sarsyun	No	Cruciferae
33	Linseed	Linum usitatissimum L.	Aalas	No	Linaceae
34	Safflower	Carthamus tinctorius L.	Kusum	No	Compositae
35	Brinjal, eggplant	Solanum melongena L.	Bhanta	Yes	Solanaceae
36	Broccoli	Brassica oleracea L. var. italica Plenck	Broccauli	Yes	Cruciferae
37	Brussels sprout	Brassica oleracea L. var. gemmiferae DC.		No	Cruciferae
38	Cabbage	Brassica oleracea L. var. capitata L.	Banda	Yes	Cruciferae

SN	English Name	Botanical Name	Nepali Name	Off season Habit	Family
39	Carrot	Daucus carota L.	Gajar	Yes	Umbelliferae
40	Cauliflower	Brassica oleracea L. var. botrytis L.	Phulgobi	Yes	Cruciferae
41	Knol-Khol	Brassica oleracea var. gongylodes L.	Ghyath gobi	Yes	Cruciferae
42	Lamb's Quarter, Chenopodium, Goosefoot, Dogtooth	Chenopodium album L.	Bethe	Yes	Chenopodiaceae
43	Goose foot	Chenopodium murale L	Kalo bhethe	Yes	Chenopodiaceae
42	Garden Cress	Lepidium sativum L.	Chamsur	Yes	Cruciferae
43	Lettuce/Curly mustard	Lactuca sativa L.	Jiriko saag	Yes	Labiatae
44	Radish	Raphaus sativus L.	Mula	Yes	Cruciferae
43	Spinach	Spinacia oleracea L.	Palungo	Yes	Chenopodiaceae
44	Turnip	Brassica rapa L.	Salgum	No	Cruciferae
45	Aniseed, Dill	Anethum graveolens L.	Sauff	No	Umbelliferae
46	Celery	Apium graveolens	Celery	No	Umbelliferae
47	Fennel	Foeniculum vulgare Mill.	Saup	No	Umbelliferae
48	Coriander	Coriandrum sativum L.	Dhania	Yes	Umbelliferae
49	Cang-kong	Ipomoea aquatica Forssk.	Karmi Sag	Yes	Convolvulaceae
50	Bishop's weed, Caraway, Ajowan, Ammi	Trachyspermum ammi (L.) Sprague	Jwano	No	Apiaceae
51	Parsley	Petroselinum crispum (Mill.) Fuss	Parsley	No	Apiaceae
52	Black cumin	Bunium persicum (Boiss.) B.Fedtsch.)	Himali Jira, Kalo Jira	No	Apiaceae
53	Chinese Leek	Allium tuberosum Rottler. ex Spreng.	Dundusaag	No	Amaryllidaceae
54	Onion	Allium cepa L.	Pyaj	yes	Amaryllidaceae
55	Cumin	Cuminum cyminum L.	Jeera	No	Apiaceae
56	Fenugreek	Trigonella foenum-graecum L.	Methi	Yes	Leguminoceae
57	Fragrant onion, leek	Allium porrum L.	Leek	Yes	Amaryllidaceae
58	Garlic	Allium sativum L.	Lasun	Yes	Amaryllidaceae
59	Nepal Aromatic Garlic	Allium hypsistum Stearn	Zimbu	No	Amaryllidaceae
60	Perilla	Perilla frutescens Brittoon	Silam	No	Lamiaceae
61	Saffron	Crocus sativus L.	Keshar	No	Iridaceae
62	Shallot	Allium ascalonicum L.	Chhyapi	Yes	Amaryllidaceae
63	Tomato	Solanum lycopersicum L.	Golvenda		Solanaceae
65	Swisschard	Beta vulgaris sub sp. vulgaris, Cicla-Group and Flavescens-Group	Swisschard	No	Amaranthaceae
66	Pea	Pisum sativum Linn.	Kerau	No	Leguminoceae
67	Asparagus	Asparagus officinalis Linn. var altilis Linn.	Kurilo	No	Liliaceae
68	Spinach	Spinacia oleracea L.	Palungo	No	Chenopodiaceae

Table 3. Lists of rainy season crops

SN	English Name	Botanical Name	Nepali Name	Off-season Habit	Family
1	Rice	Oryza sativa L	Dhan	Yes	Gramineae

SN	English Name	Botanical Name	Nepali Name	Off-season Habit	Family
2	Maize	Zea mays L	Makai	Yes	Gramineae
3	Finger millet/African millet/Bird's foot millet	Eleusine coracana (L.) Gaertn.	Kodo	No	Gramineae
4	Barnyard Millet	Echinochloa frumentacea Link	Sama	No	Gramineae
5	Foxtail Millet	Setaria italica (L.) Beauv.	Kaguno	No	Gramineae
6	Little Millet	Panicum sumatrense Roth ex Roem. and Schuit	Kutki	no	Gramineae
7	Blackgram	Vigna mungo (L.) Hepper	Mas	No	Leguminoceae
8	Jute	Corchorus capsularis L. C. olitrious L.	Jute	no	Tiliaceae
9	Sunhemp	Crotolaria juncea L.	Sanai	no	Leguminoceae
10	Cotton	Gossypium arboreum L.	Kapash	no	Malvaceae
11	Ginger	Zingiber officinale Rosc.	Aaduwa	no	Zingeberaceae
12	Pearl Millet	Pennisetum glaucum (L.) R.Br.	Bajara	no	Gramineae
13	Proso Millet	Panicum miliaceum L.	Chino	no	Gramineae
14	Paspalum	Paspalum scrobiculatum L.	Kodi	no	Gramineae
15	Cassava, Tapioca	Manihot esculenta Crantz	Simaltarul	no	Euphorbiaceae
16	Sweet potato	Ipomoea batatas Lam.	Sakharkhanda	No	Convolvulaceae
17	Groundnut/Peanut	Arachis hypogaea L.	Badam	No	Leguminoceae
18	Perilla	Perilla frutescens (L.) Britton.	Silam	no	Lamiaceae
19	Sunflower	Helianthus annus L.	Surya mukhi	no	Compositeae
20	Soybean	Glycine max (L.) Merr.	Bhattamas	No	Leguminoceae
21	Sorghum/Great millet	Sorghum bicolor (L.) Moench	Junelo	no	Gramineae
22	Pearl millet/Bajra	Pennisetum typhoides (Burm.f.) Stapf & C.E.Hubb.	Bajra	no	Gramineae
23	Little millet	Panicum miliare Lam.	Kutki sama	no	Gramineae
24	Kodo millet	Paspalum scrobiculatum L.	Kodi	no	Gramineae
25	Sesbania	Sesbania bispinosa/canabina (Jacq.) W. Wight	Dhaincha	no	Leguminoceae
26	Niger	Guizotia abyssinica Coss.	Philingo, Jhuse Til	No	Compositae
27	Sesame	Sesamum indicum L.	Til	Yes	Pedaliaceae
28	Taro	Colocasia esculenta L	Pidalu	No	Araceae
29	Yam	Diascorea alata	Taraul	No	Dioscoreaceae
30	Elephant foot yam, ole, Telagu Potato	Amorphophallus paeoniifolius (Dennst.) Nicolson),	Ole	No	Araceae

Table 4. Lists of summer season crops

SN	English Name	Botanical Name	Nepali Name	Off-season Habit	Family
1	Corn/Maize	Zea mays L.	Makai	Yes	Gramineae
2	Dent corn	Zea mays var. *indentata* (Sturtev.) L.H.Bailey	Dhate Makai	Yes	Gramineae
3	Pop corn	Zea mays var. *everta* (Sturtev.) L.H.Bailey	Murali makai	Yes	Gramineae
4	Pod corn	Zea mays var. *tunicata* A.St.Hil.	Kali Makai	Yes	Gramineae
5	Bean, French	Phaseolus vulgaris (L)	Simi	Yes	Leguminoceae

SN	English Name	Botanical Name	Nepali Name	Off-season Habit	Family
	Bean, Kidney bean				
6	Cowpea	Vigna unguiculata (L.) Walp.	Bodi	Yes	Leguminoceae
7	Double Bean	Phaseolus vulgaris L.	Asare simi	Yes	Leguminoceae
8	Sesame	Sesamum indicum L.	Til	Yes	Pedaliaceae
9	Ash Gourd, Wax Gourd	Benincasa hispida (Thunb.) Cogn.	Kuvindo	No	Cucurbitaceae
10	Balsam Apple	Momordica balsamina L.	Barela	No	Cucurbitaceae
11	Bitter Gourd	Momordica charantia L.	Tite karela	No	Cucurbitaceae
12	Bottle Gourd	Lagenaria siceraria (Molina) Standl.	Lauka	Yes	Cucurbitaceae
13	Cucumber	Cucumus sativus L.	Kankro	Yes	Cucurbitaceae
14	Drumstick	Moringa oleifera Lam.	Sajiwan, Sital Chini	No	Moringaceae
15	Okra, Ladys finger	Abelmoschus esculentus (L.) Moench	Bindi	No	Malvaceae
16	Pointed Gourd	Trichosanthes dioica Roxb.	Parbal	Yes	Cucurbitaceae
17	Summer squash	Cucurbita pepo L.	Deshi pharsi	Yes	Cucurbitaceae
18	Pumpkin	Cucurbita maxima	Parshi	Yes	Cucurbitaceae
19	Ridge Gourd	Luffa acutangula (L.) Roxb.	Pate Ghiraula	Yes	Cucurbitaceae
20	Snake Gourd	Trichosanthes anguina L.	Chichindo	Yes	Cucurbitaceae
21	Sponge Gourd	Luffa cylindrica (L.) Roem	Ghiraula	Yes	Cucurbitaceae
22	Water Cress	Nasturtium officinale R.Br.	Simsag		Brassicaceae
23	Greater Yam, White	Dioscorea alata L.	Tarul, Ghar		Dioscoreaceae
24	Turmeric	Corcuma longa L.	Besar		Zingiberaceae
25	Tobacco	Nicotiana tabacum L.	Surti		Solanaceae
26	Cotton	Gossypium arboreum L.	Kapash		Malvaceae
27	Mungbean	*Vigna radiate* (L.) R. Wilczek	Mung mass		Leguminoceae
28	Amaranthus	Amaranthus tricolor L	Latte, Marshe, Lude		Amarantaceae
29	Amaranth (Red)	Amaranthus tricolor L.			Amaranthaceae
30	Amaranth (grain)	Amaranthus cruentus L.			Amarantaceae
31	Jute	Chorcorus capsularis L.			Amaranthaceae
32	Okra	Abelmoschus esculentus (L.) Moench			Malvaceae
34	Chayote	Sechium edule (Jacq.) Sw.	Iskush	yes	Cucurbitaceae
35	Arabica coffee	Coffea arabica L.			Rubiaceae
36	Robusta coffee	C. robusta L.Linden			Rubiaceae
37	Black Pepper, Pepper	Piper nigrum L.	Marich	No	Piperaceae
38	Bell (Sweet) Pepper	Capsicum annum L. var. annum	Bhede Khursani	Yes	Solanaceae
39	Chilly	Capsicum frutescens L.	Khursani	yes	Solanaceae
40	Cinnamon	Cinnamomum zeylanicum Nees	Dalchini	Yes	Lauraceae
41	Cinnamon, Bay Leaf, cassia	Cinnamomum tamala (Buch-Ham.) Nees & Eberm.	Teejpatta	No	Lauraceae
42	Large/ greater Cardamom	Amomum aromaticum Roxb.	Alaichi	No	Zigiberaceae
43	Long Pepper	Piper longum L.	Pipala	No	Piperaceae

SN	English Name	Botanical Name	Nepali Name	Off-season Habit	Family
44	Mint (Field, Garden)	Mentha longifolia	Pudina		Lamiaceae
45	Hot Pepper	Capsicum frutescens L	Piro Khursani	Yes	Solanaceae
46	French bean	Phaseolus vulgaris Linn.	Ghui Simi, Dal simi, Asare simi	Yes	Leguminoceae
47	Peruvian ground apple, Yakon	(Smallanthus sonchifolius (Poepp.) H.Rob.)	Bhui Shyau	No	Asteraceae

Table 5. Lists of winter fruits

SN	Nepali Name	English Name	Botanical Name
1	Jangalisyau	Wild apple	Malusbaccata (L.) Borkh. Var. himalaica (maxim.)
2	Jangalikhurpani	Wild apricot	Prunuscornuta (Wall. Ex. Royle) Stud.
3	Paiyun	Wild cherry	Prunusnepalensis (Ser.) Stud.
4	Jangaliangur	Wild grape	VitislanataRoxb.
5	Jangalinaspati	Wild pear	PyruspashiaBhcu. Ex. D. Don
6	Bhui aiselu	Wild strawberry	Duchesniaindica (Andr.)
7	Aaru	Peach	Prunuspersica L. Batsch.
8	Aarubakhada	Plum	PrunusdomesticaL.
9	Ainselu	Raspberry	RubusellipticusSm. Rubusobcordatus (Franch.) Thuan
10	Amala	Gooseberry	Phyllanthus emblica L., Emblica officinalis Gaertn.
11	Amaro	Hog-plum	Spondiaspinnata (Linn. F.) Kurz., S. mangifera Willd., S. acuminate Roxb. Non Gamble.
12	Angeri, Kali chulesi or kali angeri		Osbeckia nepalensi sHook.
13	Archal		Aporusaoctandra (Buch. Ham. Ex D. Don)
15	Satibayar		Rhusparvi flora Roxb.
16	Ban timilo		Ficusfoveolata Wall.
17	Bhogate	Pummelo	Citrus maxima Merr.
18	Bhuinkaphal	Local/wild strawberry	Fragariavesca Walter
19	Bimiro	Citron	Citrus medica L.
20	Chaaksi	Sweet lime	Citrus limettioidesTan.
21	Dale chuk	Seabuckthorn	Hippophae tibetana Schltdl.
22	Dante Okhar	Thin shelled walnut	Juglans regia L.
23	Haluwabed	Persimmon	Diospyros kakiL.f.
24	Kafal	Bay berry	Myrica esculenta Buch.-Ham.exD.Don
25	Katus	Chestnut	Castanopsis hystrix A. DC., C. indica Roxb., Castanea crenata Siebold & Zucc.
26	Kimbu	Mulberry	Morus alba L., M. nigra L., M. rubra L.
27	Khurpaani	Apricot	Prunus armeniaca L.
28	Lapsi	Nepalese hog plum	Choerospondias axillaris (Roxb.) B.L.Burtt&A.W.Hill
29	Laukaat	Loquat	Eriobotrya japonica (Thunb.) Lindl.
30	Muntala	Kumquat	Fortunella japonica Swingle
31	Plum		Prunus species
32	Kusum		Schleichera oleosa (Lour.) Oken
33	Khanayo	Tree fig	Ficus semicordata Buch Ham (F. cunia Ham.)
34	Ghangaroo	Nepalese firethorn	Pyracantha crenulata Roem.

SN	Nepali Name	English Name	Botanical Name
35	Kimbu	Mulberry	Morus alba, M. serrate, M. indica Linn.
36	Naspati	Pear	Pyrus pyrifolia (Burm.) Nak., P. communis L.
37	Seetaphal or Sarifaa	Custard apple	Annona squamosal L.
38	Nibuwa	Lemon	Citrus limon (L.) Osbeck
39	Bhakiamilo		Rhus javanica Linn.
40	Painyu	Himalayan wild cherry	Prunus cerasoide sD. Don.
41	Mel or mayal	Wild pear	Pyrus pashia Linnaeus
42	Jamuna		Syzygium cumini (L.)
43	Sankhatro	Grapefruit	Citrus paradise Macfad.
44	Tindu	Wild Persimonn	Diospyros malabarica (Desr.) Kostel.
45	Thekifal	Kiwifruit, Chinese gooseberry	Actinidia deliciosa C.F.Liang&A.R.Ferguson
46	Kalo Jamun	Black plum/ Java plum /Jambol	Eugenia jambolana Lamk.
47	Kafal	Bay berry	Myrica esculenta Buch.-Ham.ex D.Don.
48	Amala	Wild gooseberries	Emblica officinalis Gaerth.
49	Kusum	Ceylon oak	Schleichera oleosa (Lour.) Merr.
50	Hade bayar	Bead plum	Zizyphus incurva Roxb.
51	Sati bayar	Nepali Sumac	Rhus parviflora Roxb.
52	Jangali bayar	Zigyphus (wild)	Zigyphus mauritiana Lamk.
53	Sankhatro	Grapefruit	Citrus paradisica Macf.
54	Bimiro	Citron	Citrus medica Linn.
55	Nilo Ainselu	Blue berries	Vaccinium uliginosumLinn.
56	Pahelo Ainselu	Yellow raspberry	Rubus ellipticus Smith.
57	Jangali Kera	Wild banana	Musa acuminata Colla.
58	Bhogate	Pummelo	Citrus grandis (Linn.) Osbeck.
59	Chiuri	Butter tree	Bassia butyracea Roxb.

Table 6. Lists of summer fruits

SN	Nepali Name	English Name	Botanical Name
1	Jangalikera	Wild banana	Musa nepalensis Wall.
2	Jangaliaamp	Wild mango	Magnifera sylvestica
3	Ambak, Belauti	Guava	Psidium guajava L.
4	Angeri, Kali chulesi or kali angeri		Osbeckia nepalensis Hook.
5	Badahar	Monkey Jack	Artocarpus lakoocha Wall. ex Roxb
6	Bayar		Zizyphus jujube Mill.
7	Hade Bayar		Zizyphus incurve Roxb.
8	Satibayar		Rhus parviflora Roxb.
9	Bael	Wood apple	Aegle marmelos (L.)
10	Ban timilo		Ficus foveolata Wall.
11	Chiuri	Bassia	Bassia/Madhuca/ Diploknema butyracea (Roxb.) H.J.Lam
12	Dante Okhar	Thin shelled walnut	Juglans regia L.
13	Emli	Tamarind	Tamarindus indica L.
14	Ghumauro-kanda		Polygonum perfoliatum Linn.
15	Harro		Terminalia chebula Retz.
16	Nivaro	Fig	Ficus carica L.
17	Muntala	Kumquat	Fortunella japonica Swingle
18	Seetaphal or Sarifaa	Custard apple	Annona squamosal L.
19	Jamuna		Syzygium cumini (L.)
20	Gulab-Jamun	Rose apple	Syzygium jambos L. (Alston)

SN	Nepali Name	English Name	Botanical Name
21	Kusum	Ceylon oak	Schleichera oleosa (Lour.) Merr.
22	Hade bayar	Bead plum	Zizyphus incurva Roxb.
23	Sati bayar	Nepali Sumac	Rhus parviflora Roxb.
24	Jangali bayar	Zigyphus (wild)	Zigyphus mauritiana Lamk.

Classification based on Growing Degree Days (GDU or GDD)

Research has shown that crop development and the time from planting to flowering and/or maturity is more closely correlated with temperature than with the number of days after planting. Growing degree units (GDU), also known as growing degree days (GDD), is used to relate temperature to crop development. GDUs are accumulated when the mean daily temperature exceeds a threshold identified for a crop. For example, wheat has a threshold temperature of 7.5°C. If the mean daily temperature for a day is 8°C, one GDU will accumulate in that day. If the next day's mean temperature is equal to 10°C, then 5 GDUs will accumulate for the day and 6 GDUs will have accumulated for the two-day period. The basic equation for computing accumulated GDU is sum of mean temperature –base temperature (Tb)

$$GDU = \frac{\sum \ Tmax}{2} - \frac{Tmin}{2} - Tb$$

where T_{max} is the maximum daily temperature, T_{min} is the minimum daily temperature, and T_b is the threshold temperature. Base temperature refers to the temperature at which growth of plant is zero.

If the daily average temperature, computed as the sum of the maximum and minimum temperatures divided by 2 minus the base temperature, is less than zero, the GDU accumulation for that day is zero. Suppose mean temperature is 32°C and base temperature is 4°C, the accumulated amount of heat will be 32-4 = 28°C/day. This temperature is added daily till plant ends its life cycle. Cereal crops require 990-1200 heat units. For cool-season crops, such as cereal crops and most C3 crops, the base temperature is 7°C. For warm season and most C4 crops, a modified GDU method is used. The basic equation is the same but if the minimum temperature is lower than the base temperature, then the day's minimum temperature is set equal to the base temperature, usually 10°C. The maximum daily temperature is also modified if the daily maximum temperature exceeds 30°C.

Table 7. Lists of crops which requires a comparatively low temperature range, 7-13°C

SN	Agronomic Crops	Horticultural Crops		
		Vegetables	Fruits	Flower
1	Wheat	Asparagus	Apple	Aster
2	Barley	Spinach	Pear	Carnation
3		Lettuce	Plum	Cineraria
4		Cabbage	Cherry	Garanium
5		Radish	Grape	Petunia
6		Turnip	Strawberry	Snapdragon
7		Carrot	Blueberry	Verbena
8		Pea	Raspberry	Vinca
9		Potato	Apricot	
10			Almond	
11			Quince	
12			Apricot	
13			Currant	

Table 8. Lists of crops which require a comparatively moderately high temperature range, 13-18° C

SN	Agronomic Crops	Horticultural Crops		
		Vegetables	Fruits	Flower
1	Rice	Tomato	Peach	Asparagus plumosus
2	Maize	Capsicum	Plum	Africon violet
3	Sorghum	Eggplant	Fig	Azalea
4		Cucurbits	Persimmon	Begonia
5		Bean	Pecan	Cattleya
6			Grape	Orchid
7			Avocado	Coleus
8			Olive	Gardenia
9			Guava	Hydrangea
10				Kalanchoe
11				Lily
12				Poinsettia
13				Rose

Table 9. Lists of crops which require a comparatively high temperature range, 18-24°C

SN	Agronomic Crops	Horticultural Crops		
		Vegetables	Fruits	Flowers
1	Maize	Sweet potato	Banana	Philodendron
2		Yam	Cashew nut	Dieffenbachia
3		Cassava	Citrus	Croton
4		Chayote	Coffee	Pothos
5		Okra	Date palm	Ficus
6		Cucurbits	Fig	Peperomia
7		Sweet corn	Papaya	Dracaena
8			Pomegranate	Aspidistra
9			Litchi	Gloxinia
10			Mango	
11			Pineapple	
12			Jackfruit	
13			Bael	
14			Coconut	
15			Sapota	
16			Mangosteen	

Heat unit varies with change in location or altitude. The different locations will have different heat units. A maize variety that matures in 90 days at Parwanipur condition will need 120 days to mature in Munstang. Cole crops are annual at Parwanipur and Chitwan but biennial at high altitude such as Jomsom, Mustang. A variety of wheat matures at 90 days in Rauthot at 120 days in subtropical region and at 180 days in temperate region. GDD is more reliable and accurate than mean temperature of specific crops.

Classification according to CO_2 Fixation or Photosynthetic Behavior

C3 Plants

Net assimilation rate in these plants is low (15- 40 mg CO_2/d^2) but compensation point is high (30-70 ppm). Stomata are opened during the day. Photorespiration is high in these plants. C3 plants have lower water use efficiency. The initial product of assimilation is the three 'C' compounds. The enzyme involved in the primary carboxylation is ribulose-1, biophospate carboxylose. For example, rice, soybean, wheat, barley, cotton, potato. Those C3 plants include more than 95% of the plant species on earth eg rice, wheat, oat, barley, cotton, beans, sunflower, spinach.

C4 Plants

Net assimilation rate in these plants is high (40- 80 mg CO_2/d^2) but compensation point is low (0- 15 ppm). Stomata are opened during the day. The primary product of 'C' fixation is four carbon compounds which may be malice acid or acerbic acid. The enzymes responsible for carboxylation are Phosphoenol Pyruvic acid carboxylose which has high affinity for CO_2 and capable of assimilation CO_2 event at lower concentration, photorespiration is negligible. These are said to be drought resistant & they are able to grow better even under moisture stress. C4 plants translate photosynthates rapidly, et sugarcane, sorghum, maize, sesame, amaranths, etc.

CAM Plants

CAM stands for Crassulacean Acid Metabolism, and in such plants, the stomata open at night and large amount of CO_2 is fixed as a malice acid which is stored in vacuoles. During day stomata are closed. There is no possibility of CO_2 entry. CO_2 which is stored as malice acid is broken down and released as CO_2. In these plants there is negligible transpiration. C4 and CAM plants have high water use efficiency. These are highly drought resistant eg pineapple, sisal, bryophyllum, opuntia, dragon fruit, agave, etc.

Classification of Crop Plants based on Frost Tolerant

Vegetables are classified based on ability of crops to withstand frost and cold. This classification of crop is thermo classification. Vegetable crops can be grouped as hardy, half hardy, and tender based on degree of frost tolerance. Few vegetables and fruits require cold temperature for the efficient growth and development.

Cool-season crops grow poorly in the summer heat. Cool-season crops thrive best under cool conditions. Average daily temperature is 21°C or less. Specific climate and weather may influence planting dates. The soil temperature at the depth where the seeds are planted should be at least 4.5°C. Winter crops are annual crops sown in autumn and are harvested in spring or summer. Cool-season crops come to harvest in cool weather, either in spring or fall or winter. Cool-season crops can be planted when the soil and air temperatures are cool, as low as 5°C. Mature cool-season crops can survive in temperatures near freezing without protection. Cool-season crops do not do well in the warmest summer temperatures. Beet, cabbage, cauliflower, celery, parsley, pea, radish, swiss chard, turnip require a minimum of 4°C and an optimum of 27-29°C. Cool-season vegetables grow best when the temperature highs are in the range of 21-24°C. Cool-season crops usually stop producing when daytime temperatures reach 26ºC or higher. Many cool-season crops that come to maturity before or shortly after the first frost in autumn can be protected where they are in the garden from freezing temperatures and harvested as needed throughout the winter. These plants do not continue to grow, but simply maintain and remain ready for harvest. Collards, cabbage, spinach, and kale are among the most cold-hardy vegetables. Frost on these collard leaves makes them sweeter. Hardy and semi-hardy vegetables are planted in early spring for spring harvests and again in late summer for fall harvests. Cool-season vegetables are often further categorized as hardy and half hardy.

Hardy

Hardy crops are the most cold tolerant. They can be planted 2 to 4 weeks before the date of the average last frost in spring; their seeds will germinate in cold soil and their seedlings can endure short freezes. Hardy vegetables will grow when the daytime temperature is as low as 5°C. Hardy vegetables include asparagus, broccoli, brussels sprouts, cabbage, chives, collards, corn salad, garlic, horseradish, kale, kohlrabi, leeks, mustard, onions, parsley, peas, radish, spinach, and turnips. Hardy vegetables tolerate hard frosts (usually 25 to 28°F). They are good for spring and fall gardens. The hardiest–kale, spinach, and collards–can tolerate low temperatures. All taste best when they mature in cool weather, so they are very well suited to late summer planting for fall harvests.

Half-hardy or Semi-hardy

Half hardy crops grow when the minimum temperature is between 5-10°C. They are able to tolerate light freezes ie just a few hours of frost. Half-hardy vegetables include beets, carrots, cauliflower, celery, celeriac, chard, Chinese cabbage, chicory, globe artichokes, endive, lettuce, parsnips, potatoes, and hardy herbs. Half-hardy crops may be planted as early as 2 weeks before the average last spring frost. Semi-hardy vegetables tolerate light frosts (usually 29 to 32°F) late into fall and through winter in mild climates. They are good for spring and fall gardens. Examples are beets, carrot, cauliflower, celery, Chinese cabbage, endive, Irish potatoes, lettuce and gourmet salad greens, radicchio, rutabaga, salsify and swiss chard. Swiss chard and collards taste best in the cool of spring and fall, but will hang on during summer heat, too.

Table 10. Classes of vegetables based on thermo-tolerance

SN	Cool Season Crops		Warm Season Crops	
	Hardy	Half Hardy	Tender	Very Tender
1	Asparagus	Beet	Cowpea	Cucumber
2	Broad bean	Carrot	Snap bean	Eggplant
3	Brussels sprout	Cauliflower	Soybean	Lima bean
4	Broccoli	Celery	Sweet corn	Muskmelon
5	Cabbage	Chinese cabbage	Tomato	Okra
6	Chives	Lettuce		Hot pepper
7	Garlic	Parsnip		Sweet pepper
8	Horse radish	Potato		Pumpkin
9	Kale	Amaranth		Squash
10	Kohlrabi	Fennel		Sweet potato
11	Leek	Globe artichoke		Water melon
12	Broad Leaf Mustard	Swiss chard		
13	Onion	Spinach		
14	Parsley			
16	Peas			
17	Radish			
18	Spinach			
19	Turnip			

Tender and Very-tender Crops

Tender vegetables prefer temperatures between 21-35°C and require daytime temperatures of 15°C or greater to thrive. Tender crop seedlings are best transplanted into the garden 1 to 2 weeks after the last spring frost. Tender vegetables include beans, celery, sweet corn, cucumbers, New Zealand spinach, summer squash, and tomatoes.

Very-tender vegetables require daytime temperatures consistently warmer than 13°C. A week of daytime temperatures below 55°F will likely stunt the growth of a very tender vegetable. These crops are likely to suffer in a cool breeze. They are best planted at least 3 weeks after the last frost in spring. Very-tender vegetables include lima bean, eggplant, muskmelon, okra, pepper, pumpkin, winter squash, sweet potato, and watermelon.

Off-season Crops

Off-season cultivation of vegetable refers to the production of the vegetable before or after their normal cultivation seasons of production. This can be accomplished by using the different agro-climatic conditions, adjusting the planting time, selecting and improving the varieties or by creating the controlled environment by making the plastic tunnels, polythene houses, permanent glass houses, etc.

Vegetables have a high potential for market growth not only in Nepal but also across the border in India. Of late, off-season vegetable is being recognized as a major agriculture commodity that holds the comparative advantage for export to India (HVAP 2011). Between July and September, the agro-climatic conditions in the Tarai plain of Nepal and India become unsuitable for vegetable cultivation. This is due to water logging and flooding brought about by the monsoon. The production of major vegetable crops, such as cauliflower, cabbage, and cucurbits is badly affected by these conditions.

However, Nepal's geographic features are varied. There are hilly and mountainous regions as well. These conditions are suitable for many kinds of vegetables to be grown even during the rainy summer, right up to the autumn. Off-season vegetables fetch more prices compared to their normal season counterparts. However, farmers are required to put extra effort to save the crops from adverse weather, insect pests and diseases, besides also require careful tending. Such farming requires polyhouse technology, drip irrigation and training to farmers.

The majority of the mid-hill farmers rarely have a balanced diet, due to the lack of vegetables. Vegetables are rare in the period from January to June, when cold temperatures and little rainfall make production difficult. Techniques that overcome these conditions can help provide food and income during the time of the year when they are especially needed. Crops and growing season are given below.

Common off-season vegetables in the hills of Nepal

Off-seasons vegetable	**Growing season**
Cabbage	Year round
Cauliflower	January-March
Tomato	Year round
Radish	January-March
Brinjal	Year round
Summer squash	March-May
Cucumber	April-November
Carrot	January-March
Swiss Chard	Year round

Fall Vegetables vs Summer Vegetables

Fall season vegetables are also known as cool season vegetable. These are grown for their leaves, stalks, bulbs and roots as opposed to their fruits eg brocauli, cauliflower, cabbage, kale, collards, Asian greens, carrots, beet, parsnip, celery, fennel, onion, garlic, shallot, chive, leeks. Fall vegetables grow best in the cooler temperature of spring and fall. They thrive in temperature of 10-22°C as long as they have plenty of water. In warm-winter climates, fall vegetables planted in early September grow slowly through the winter months, then size up quickly as soon as the weather warms up in the spring.

Basic Growth Pattern for Fall Vegetables

Most cool season vegetables follow the same growth pattern. After germination, carrots, spinach and broccoli form a low rosette of leaves radiating out from the center. As long as conditions are good – steady water, decent soil, temperatures in the 16 to 23°C, fall vegetables accumulate and store nutrients. Fall vegetables need steady, continuous growth. Anything that disrupts this growth like spring heat wave or drought causes them to bolt or send up a flower stalk.

Cold tolerance is the bright line dividing fall vegetables (broccoli, spinach, and lettuce) from summer vegetables (tomatoes, melons, and cucumbers). But it's not the only line. Summer vegetables are

actually fruit, not vegetables although some (like potatoes and sweet potatoes) are grown for their fleshy tubers. Most of them evolved in the tropics. Unlike cool-season vegetables, frost kills them. They have to be planted after the last frost date in the spring, and ripen fruit before the first frost of the fall.

Summer vegetable needs higher temperatures for seeds to germinate, and summer heat to flower and set fruit. Nightshades (tomatoes, peppers, eggplants, and potatoes), cucurbits (cucumbers, melons, squash), beans, and corn fall into this group. Most summer vegetables need a long growing season to produce the most fruit. Summer vegetables are heavy feeders and need supplemental nitrogen, especially during the first part of the growing season. However, too much nitrogen (especially late in the season) results in a lot of leaves and less fruit. Although they can be grown in the same soil as cool-season vegetables, summer vegetables prefer a slightly more acidic soil (pH 5.5-6.0), and they require higher levels of phosphorus and potassium in the soil to flower and set fruit. The idea is to have large, robust plants with a lot of leaf area when it's hot enough for summer vegetables to flower and set fruit. The leaves produce the sugars that sweeten the fruit, and shade the fruit to prevent sunscald.

Summer Vegetables

Summer vegetables have a different watering regime than fall vegetables, and the plants may need to be staked, caged, or trellised to keep vines from breaking when they're heavy with fruit. Watering summer vegetables unlike fall vegetables, which need continuous water, summer vegetables perform better with deep, less frequent watering, especially early in the season. They still need regular water, but if the plants dry out a little between watering, it forces them to send their roots deeper into the soil, resulting in larger, stronger, more robust plants. Once the plants start setting fruit, regular watering is best. Irregular watering as fruit ripens may result in split fruit. Drip irrigation is better for summer vegetables. Plant diseases that plague many summer vegetables spread readily on wet foliage.

We can grow different winter and summer vegetables simultaneously taking advantages of altitude. In the three different physiographic regions, potato can be harvested all the year round. During winter potato comes up well in the Tarai whereas during that period mountains are full of snow but it will be too hot for potato to grow during summer in the Tarai. At this season in the hills and mountain potato thrives well and gives good production (Pun and Karmacharya 1998)

Biennial and Perennial Crops

These crops don't fall neatly into a cool or warm season category. Some crop specific planting and harvesting techniques are described below.

- Artichoke (perennial in sunset climate zones)- Plant in fall for spring harvest
- Asparagus- Plant seedlings or roots in fall or winter; early spring in cold-winter areas. Cut first spears early during year two; plants take three years or so to come into full production.
- Garlic- In mild winter areas, plant in fall; where winters are cold, plant in early spring.
- Leek- In cold winter regions, set out transplants in early spring, or sow seed in late summer for harvest the following year. In mild winter regions, set out transplants in fall.
- Parsley- In cold climates, plant in spring, after last frost. Plant in fall or early spring where winters are mild; in early fall in the low desert.
- Parsnip- In cold winter areas, sow seeds in late spring, harvest in fall. In mild winter areas, sow in fall for harvest in spring.
- Shallot- In mild climates, plant in fall to harvest green tops through winter and spring, bulbs in late spring and summer. In cold climates, plant in early spring for green shoots in summer, bulbs in autumn.

- Turnip- Winter crop in mild-winter areas. Where winters are cold, plant in early spring for summer harvest.

Classification of Flowers based on Growing Season

The various climatic changes that occur in cyclic patterns are termed as seasons. There are four general seasons occurring on Earth ie Spring, Summer, Autumn and Winter. Flowers and seasons are intimately bound to each other. Most of the flowers are season specific. Though some flowers are found throughout the year, there are some not particularly affected by changing seasons, viz Alstromeria, Anthurium, Aster, Bourvardia, Carnation, Chrysanthemum, Gerbera Daisy, Lily, Asiatic Lily.

Spring Flowers

Spring time is the time of growth and renewal of new plant and animal life. Spring comes at different times in the North and South Hemispheres. Spring time in the Northern hemisphere is between March - May, and between September - November in the Southern hemisphere. Most flowering plants bloom during spring time. Therefore, flowers that bloom only during spring, spring flowers, bloom at different times in the two hemispheres. Examples are Agapanthus, Amaryllis, Anemone, Apple blossom, Bird of Paradise, Brodea, Calla lily, Cherry Blossom, Corn flower, Cosmose, Dahlia, Delphinium, Delwood, Forsythia, Freesia, Gardenia, Heather, Helleborus, Hollyhock, Hyacinth, Larkspur, Casa Blanca Lily, Gloriosa Lily, Stargazer, Liatrus, Lilac, Lisianthus, Narcissus, Orchid, Peach blossom, Peony, Phlox, Poppy, Protea, Pussy willow, Ranunculus, Rose, Seeded Eucalyptus, Solidago, Statice, Stephanotis, Stock, Sweet Pea, Tulip, Viburnum, Wax flower, Zinnia.

Summer Flowers

Summer is the time of hot and warm weather. Floral growth is best in the summer season. The Northern hemisphere experiences summer during June, July, and August, while in the Southern hemisphere, during December - February. Summer is the season of the midnight sun in the North pole and Ice land. Examples are Alchemilla, Allium, Alstromeria, Amaranthus, Baby's Breath , Bird of Paradise, Calla lily, Campanula, Carnation, Chrysanthemum, Cockscomb, Cosmos, Dahlia, Delphinium, Dianthus, Didiscus, Euphorbia, Foxglove, Freesia, Gardenia, Genista, Ginger, Gladiolus, Hallaconia, Heather, Hydrangea, Hypericum, Iris, Kangaroo paw, Liatrus, Lilac, Casa Blanca Lily, Gloriosa Lily, Star Gazer, Lisianthus.

Autumn Flowers

Autumn is the season of the primary harvest. Autumn falls during September - November in the Northern hemisphere, and during September - November in the Southern hemisphere. Crops are harvested during Autumn. Beautiful leaves that change color are at their best.

Examples are Acashia, Allium, Alstromeria, Amaranthus, Anemone, Baby's Breath, Bittersweet, Carnation, China berry, Chrysanthemum, Cockscomb, Cosmos, Echinops, Freesia, Gerbera Daisy, Gladiolus, Hypericum, Iris, Juniper, Kangaroo paw, Kalancheo, Liatrus, Lily, Asiatic, Lily, Gloriosa, Misty Blue, Orchid, Pepper berry, Protea, Queen Ann's Lace, Quince, Rover, Roses, Rowen berry, Salvia, Solidago, Statice, Star of Bethlehem, Sunflower, Yarrow, Zinnia.

Winter Flowers

Winter is the season of cold weather. The season occurs during December - February in the Northern hemisphere. In the Southern hemisphere winter occurs during June - August. Examples are Acashia, Alstromeria, Amaryllis, Carnation, Chrysanthemums, Cyclamen, Evergreens, Gerbera Daisy, Ginger, Helleborus, Holly berry, Lily, Asiatic Lily, Casa Blanca Lily, Narcissus, Orchid, pandy, Pepper berry, Phlox, Protea, Queen Ann's Lace, Roses, Star of Bethlehem, Statice.

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